

Depth-Width Trade-offs in Plain Convolutional Neural Networks: A Controlled Study Under Fixed Parameter Budget

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Abstract

A fundamental question in convolutional neural network design is how to optimally allocate a fixed parameter budget between model depth and width. We hypothesise that deeper models better capture hierarchical compositional features — enabling fine-grained discrimination through successive abstraction — while wider models encode richer representations at each level through broader channel co-occurrence. Under this hypothesis, an optimal depth-to-width ratio should exist, and both extremes should underperform balanced configurations. We test this hypothesis systematically on CIFAR-100 by constructing a family of six VGG-style CNNs spanning depths from 4 to 28 layers, with widths adjusted to maintain a constant $\sim 1.5M$ parameter budget. Our experiments confirm the central prediction: a clear inverted-U accuracy curve peaks at depth 12 (74.01% top-1), with a stable optimal plateau spanning depths 12–20 and both extremes underperforming. Gradient flow analysis reveals the mechanistic basis: very shallow networks lack hierarchical capacity while very deep narrow networks suffer from suppressed gradient magnitudes throughout training. Critically, per-superclass analysis provides the most direct evidence for the hierarchical features hypothesis — depth advantages are largest in fine-grained biological categories (reptiles, people, large animals) where intra-class discrimination demands compositional feature hierarchies, while width maintains a slight advantage in texture-dominated categories such as trees. Image edge complexity, measured by Laplacian variance, does not significantly modulate the depth-width preference at 32×32 resolution. Together, these results provide both empirical confirmation and mechanistic insight into the depth-width trade-off under resource constraints.

Keywords: convolutional neural networks, depth-width trade-off, architecture design, fixed parameter budget, CIFAR-100, gradient flow, image classification, model scaling

1. Introduction

We believe the architecture of a convolutional neural network (CNN) determines how it allocates its representational capacity across layers and channels. Two of the most fundamental design axes are *depth* — the number of convolutional layers — and *width* — the number of channels per layer. We believe these axes have distinct theoretical roles: depth enables hierarchical feature composition, transforming low-level edge patterns into high-level semantic representations through successive non-linear transformations; width expands the representational vocabulary at each level, allowing richer co-occurrence statistics to be captured simultaneously.

Despite their conceptual distinctness, the practical consequences of prioritising depth over width — or vice versa — under a fixed total parameter budget remain incompletely understood. Prior work has explored these dimensions largely in isolation: VGGNet [10] demonstrated the power of depth; Wide Residual Networks [12] showed that width can be equally effective; and EfficientNet [11] proposed compound scaling that adjusts both simultaneously. However, these studies typically vary one dimension while changing overall capacity, or rely on skip connections that substantially alter gradient flow dynamics and confound the attribution of performance to depth vs. width.

Hypothesis. We advance the following central hypothesis: *under a fixed parameter budget, an optimal depth-to-width ratio exists, and both extremes — very shallow-wide and very deep-narrow configurations — will underperform balanced architectures.* This hypothesis follows from two theoretical premises: (i) depth is necessary to build compositional hierarchies of features, so insufficient depth limits discriminative capacity regardless of channel count; and (ii) width provides the representational vocabulary at each level, so insufficient width becomes a bottleneck that depth alone cannot compensate, particularly for fine-grained intra-class discrimination. We further predict that the depth advantage should be most pronounced in semantically complex categories requiring hierarchical fea-

ture composition — such as fine-grained biological classes — while width may provide a comparative advantage in categories characterised by distinctive large-scale texture or colour patterns.

In this work we test these hypotheses through a controlled experimental design on CIFAR-100 [6]. We construct a family of six plain VGG-style CNNs whose parameter counts remain within $\pm 5\%$ of a shared $\sim 1.5\text{M}$ budget while depth varies from 4 to 28 layers and width adjusts inversely. Crucially, no skip connections are used, channels are held constant across all three spatial stages (no channel doubling), and all other hyperparameters are fixed identically. This enables a direct, clean comparison attributable solely to the depth-width axis, free from architectural confounds.

Our investigation goes beyond classification accuracy to include convergence dynamics, gradient flow analysis, image complexity stratification, and per-superclass accuracy — providing both empirical evidence and mechanistic insight into the mechanisms underlying the trade-off.

Contributions.

- A controlled family of six CNNs with strictly matched $\sim 1.5\text{M}$ parameter budgets spanning a wide range of depth-width configurations (§3).
- Empirical confirmation of the central hypothesis: an inverted-U accuracy curve on CIFAR-100 peaking at depth 12, with a stable optimal plateau at depths 12–20 (§4).
- Gradient flow analysis revealing that deep-narrow networks (depth 28) experience systematic gradient magnitude suppression, providing a mechanistic explanation for their convergence gap despite Batch Normalization (§5).
- Semantic analysis directly supporting the hierarchical features prediction: depth advantages are concentrated in fine-grained biological and structural categories, while the complexity-tier analysis shows that raw pixel-level edge density does not modulate the trade-off at 32×32 resolution (§5).

2. Related Work

2.1. Depth in CNNs

VGGNet [10] established that stacking many 3×3 convolutions consistently improves image classification, demonstrating the value of depth for building multi-scale representations. ResNet [2] extended this to hundreds of layers via residual connections, which provide shortcut gradient paths and largely resolve the vanishing gradient problem. While ResNets have become the de-facto standard backbone, their skip connections fundamentally alter both

the optimization landscape and the attribution of performance to specific depths, making them less suitable for studying the depth-width trade-off in isolation.

2.2. Width in CNNs

Wide Residual Networks (WRN) [12] showed that widening ResNets with a widening factor k (up to $k=10$) while reducing depth achieves strong performance with faster training. Their results challenged the prevailing assumption that depth is always preferable, showing that width can compensate for reduced depth when total capacity is held roughly constant. MobileNets [4] introduced width multipliers as a systematic efficiency mechanism, decoupling capacity from depth, but primarily in the context of reducing computation rather than studying the depth-width interaction.

2.3. Compound Scaling

EfficientNet [11] jointly scales depth, width, and input resolution using coefficients derived from neural architecture search on a small baseline model. While comprehensive, this approach requires a separate search phase and treats the three dimensions as co-optimised rather than isolating their individual contributions. Our work complements this by holding resolution and total capacity fixed, enabling a principled analysis of the depth-width axis alone.

2.4. Gradient Flow and Optimization

The vanishing gradient problem [3] poses a fundamental challenge for deep networks, as gradients diminish exponentially through many non-linear layers during backpropagation. Batch Normalization (BN) [5] largely mitigates this by standardising layer inputs, stabilising gradient magnitudes and enabling training of much deeper networks. Nevertheless, prior work [1] notes that BN does not guarantee uniformly strong gradient flow across all depths; our gradient analysis examines this empirically for plain CNNs without skip connections. Visualisation of loss landscapes [7] has shown that deeper networks without skip connections tend to have sharper, more chaotic loss surfaces, consistent with the optimization difficulties we observe in our deepest models.

2.5. Representation Differences Between Deep and Wide Networks

Nguyen *et al.* [9] investigated how depth and width affect learned internal representations using centred kernel alignment, finding that wider networks develop a distinctive “block structure” in which many consecutive layers converge to nearly identical representations. Their work is complementary to ours: where they study *what* is learned differently, we study *how much* is learned correctly — as measured by downstream classification accuracy, gradient

Table 1. CNN architecture family. All models share a ~ 1.5 M parameter budget. Stage depths denote layers per stage (S1, S2, S3).

Model	Depth	Width	Params	ΔP^*	Stages
M1	4	233	1,497,358	-0.18%	(1,1,2)
M2	8	153	1,496,746	-0.22%	(2,2,4)
M3	12	122	1,492,038	-0.53%	(4,4,4)
M4	16	105	1,505,170	$+0.34\%$	(5,5,6)
M5	20	93	1,494,610	-0.36%	(6,6,8)
M6	28	78	1,492,786	-0.48%	(9,9,10)

dynamics, and semantic category performance — under a strict parameter budget constraint.

Positioning. Our work is most closely related to WRN [12] and Nguyen *et al.* [9], but differs in three key respects: (i) we use plain (non-residual) convolutions to avoid confounding from skip connections; (ii) we fix the total parameter budget rather than fixing depth and scaling width; and (iii) we extend beyond accuracy to include gradient flow and semantic category analysis.

3. Method

3.1. Architecture Design

We adopt a VGG-style [10] template: sequences of 3×3 Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU blocks organised into three spatial stages, separated by 2×2 max-pooling and followed by global average pooling and a linear classifier. Two design choices are critical for a controlled study:

Constant width across stages. Unlike standard VGG-style networks that double channels at each pooling stage, we hold width W constant across all stages. This ensures that depth D and width W are the *sole* varying dimensions, with no confound from channel-size progression.

Fixed parameter budget. For a model with D convolutional layers (no bias) and uniform width W , the total trainable parameter count is:

$$P = 9 \cdot 3 \cdot W + (D - 1) \cdot 9W^2 + 2DW + 100W + 100, \quad (1)$$

where the terms represent the input convolution ($3 \rightarrow W$), subsequent $W \rightarrow W$ convolutions, BatchNorm parameters, classifier weights, and classifier bias respectively. Given a target budget $P^* = 1.5 \times 10^6$ and depth D , the optimal width W is obtained by solving the quadratic in Eq. (1), yielding the six (depth, width) pairs shown in Table 1. All models are within $\pm 0.53\%$ of P^* .

Depth distribution. Layers are distributed across the three stages as $\lfloor D/3 \rfloor : \lfloor D/3 \rfloor : D - 2\lfloor D/3 \rfloor$, with a minimum of one layer per stage. All weights are initialised with Kaiming normal initialisation [2].

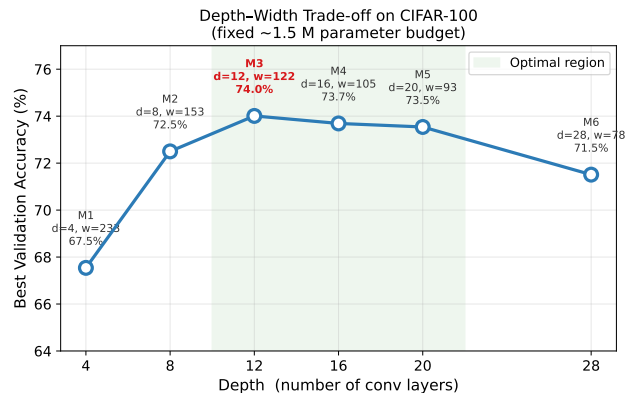


Figure 1. Best validation accuracy vs. depth for all six models. The shaded region marks the optimal plateau (depths 12–20). M3 achieves peak performance at 74.01%.

3.2. Training Protocol

All six models are trained with *identical* hyperparameters to ensure fair comparison. We use SGD with momentum 0.9 and weight decay 5×10^{-4} , an initial learning rate of 0.1 decayed by cosine annealing [8] over 200 epochs, and a batch size of 256 on an NVIDIA L4 GPU.

Data augmentation consists of random crop (32×32 with padding 4) and random horizontal flip, followed by per-channel normalisation using CIFAR-100 statistics ($\mu = (0.507, 0.487, 0.441)$, $\sigma = (0.268, 0.257, 0.276)$). No additional augmentation (Mixup, CutMix, AutoAugment) is applied, to isolate the architectural effect.

The best validation checkpoint (highest top-1 accuracy) is saved separately for each model for the analyses in §5.

4. Experiments

4.1. Dataset

CIFAR-100 [6] contains 60,000 colour images of size 32×32 across 100 fine-grained categories, organised into 20 semantic superclasses. Each class has 500 training and 100 test images. We report top-1 and top-5 accuracy on the 10,000-image test set.

4.2. Main Results

Table 2 reports the best validation accuracy and corresponding epoch statistics for each model. Figure 1 visualises the accuracy as a function of depth.

The results reveal a clear *inverted-U* pattern. Accuracy rises monotonically from M1 (67.54%) to M3 (74.01%), then declines through M4 and M5 before dropping more sharply at M6 (71.51%). Three observations are noteworthy.

The optimal plateau. M3, M4, and M5 form a stable plateau with a span of only 0.47% (73.54–74.01%). This

Table 2. Main results on CIFAR-100. Top-1 and Top-5 accuracy (%) are reported at the best validation epoch. Train Acc. is the training accuracy at the same epoch. Best model in **bold**.

Model	Depth	Best Ep.	Top-1	Top-5	Tr. Acc.
M1	4	184	67.54	89.55	98.71
M2	8	197	72.50	91.64	99.95
M3	12	190	74.01	92.10	99.92
M4	16	183	73.69	92.47	99.81
M5	20	194	73.54	92.05	99.67
M6	28	191	71.51	91.62	98.69

suggests that the optimal depth-to-width ratio occupies a range rather than a precise point, providing robustness to moderate architectural variation around the optimum.

The cost of extreme shallowness. M1 (depth 4, width 233) underperforms the best model by 6.5%. Despite its large channel count, four convolutional layers are insufficient to build the hierarchical feature representations needed for 100-class fine-grained classification on 32×32 images.

The cost of extreme depth. M6 (depth 28, width 78) underperforms M3 by 2.5% and even lags behind M2 (depth 8) by 1.0%. Notably, M6 also achieves only 98.7% training accuracy — the lowest among all models — despite 200 training epochs. As depth increases to 28 layers, the 78-channel width becomes a bottleneck both for representational capacity and, as we show in §5.1, for gradient propagation.

4.3. Convergence Analysis

Figure 2 shows validation accuracy and training loss over 200 epochs. All models converge smoothly without instability, confirming that the cosine annealing schedule and Batch Normalization provide adequate optimisation conditions.

However, the training loss curves (right panel, log scale) reveal a meaningful separation. The mid-depth models M2–M5 reach training losses near 10^{-2} , while M1 stalls at approximately 10^{-1} and M6 sits between the two groups at $\sim 6 \times 10^{-2}$. This gap in training loss — despite all models having the same parameter count — indicates that depth and width interact with the optimisation dynamics independently of raw capacity. M1’s high training loss points to a representational capacity limitation that width alone cannot resolve, while M6’s intermediate loss is consistent with a gradient propagation issue that we investigate next.

5. Analysis

5.1. Gradient Flow

To investigate the optimisation dynamics underlying the convergence gap, we record the L2 norm of gradients for every convolutional layer at each training epoch and report the average over all 200 epochs.

Figure 3 reveals three patterns. First, M1 (depth 4) exhibits markedly higher gradient norms (up to 2.0) throughout its four layers. With only four layers sharing the full optimisation burden, each layer receives stronger gradient signals per unit depth — consistent with M1’s fast initial learning but limited asymptotic accuracy. Second, the plateau models M3–M5 maintain moderate, consistent gradient norms in the range 0.5–1.0, a regime associated with stable and effective weight updates. Third, M6’s gradient norms are systematically the *lowest* across the entire depth, particularly in early layers.

This pattern warrants careful interpretation. Batch Normalization [5] prevents the exponential decay characteristic of classical vanishing gradients: all models remain within approximately one order of magnitude on the log scale. Nevertheless, the progressive suppression of gradient magnitudes throughout M6’s 28-layer stack means that early convolutional layers receive weaker update signals throughout training. This is mechanistically consistent with M6 failing to converge as effectively as mid-depth models (98.7% vs. 99.7–99.9% for M2–M4, and 99.7% for M5) and provides a gradient-level explanation for the accuracy decline at extreme depth.

5.2. Image Complexity Analysis

To test whether image edge complexity interacts with the depth-width trade-off, we score each of the 10,000 CIFAR-100 test images using Laplacian variance — a standard proxy for image sharpness and edge density, computed as the variance of the image convolved with a Laplacian kernel. Images are ranked and divided into three equal-frequency tiers: Low ($\sim 3,333$ images with the lowest edge density), Medium, and High. Per-tier accuracy is evaluated on each model’s best-epoch checkpoint.

Figure 4 and Table 3 show the results. The High–Low accuracy gap is consistently small across all models, ranging from 0.77% (M5) to 2.24% (M3). Critically, *no crossover is observed*: the ranking among models is preserved across all three tiers, and no individual model gains a relative advantage on either extreme of the complexity spectrum.

This constitutes a **negative finding**: at 32×32 resolution, pixel-level edge density does not modulate the depth-width preference. Architectures that perform better overall perform better in all tiers. The most plausible explanation is that the image statistics differentiating the tiers — fine-grained variations in edge density — are insufficient to engage the specific hierarchical vs. co-occurrence representations that distinguish deep-narrow from shallow-wide networks at this resolution.

5.3. Semantic Category Analysis

CIFAR-100’s 20 semantic superclasses allow a richer decomposition of performance differences and provide the

Convergence Behaviour — All 6 Models

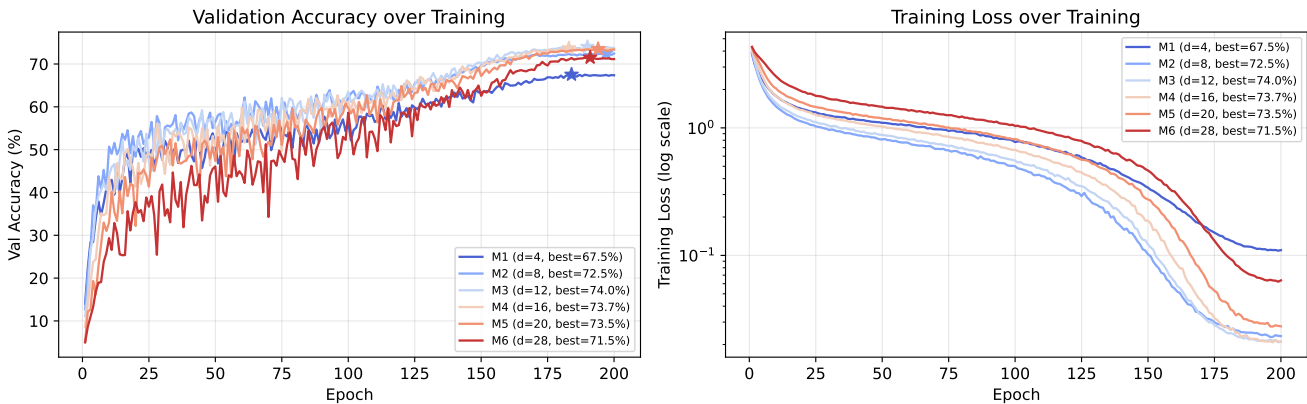


Figure 2. Convergence curves for all six models over 200 epochs. **Left:** Validation accuracy; stars mark each model’s best epoch. **Right:** Training loss on a log scale, highlighting the divergence between the extreme models (M1, M6) and the mid-depth models (M2–M5).

Gradient Flow: Mean Conv Layer Gradient Norm Across Training

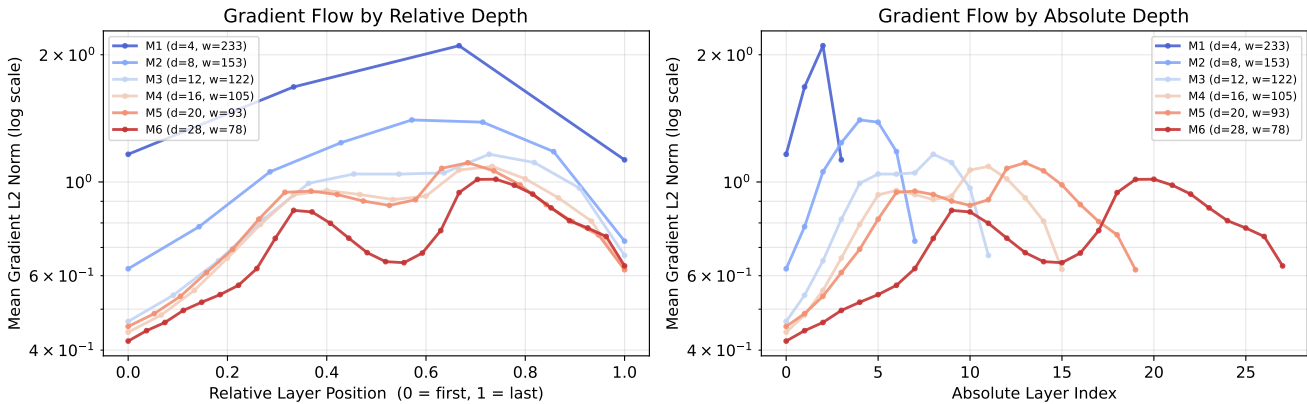


Figure 3. Mean gradient L2 norm per convolutional layer, averaged over all 200 training epochs. **Left:** Normalised layer position (0 = first, 1 = last). **Right:** Absolute layer index. Deeper-narrower models consistently operate at lower gradient magnitudes, with M6 (depth 28) showing the most suppressed flow.

Table 3. Per-complexity-tier accuracy (%) using Laplacian variance to stratify the 10,000 CIFAR-100 test images into three equal frequency tiers.

Model	Overall	Low	Medium	High
M1	67.54	67.52	66.55	68.56
M2	72.50	72.05	71.83	73.63
M3	74.01	73.31	73.18	75.55
M4	73.69	73.31	72.76	75.01
M5	73.54	73.64	72.58	74.41
M6	71.51	71.57	70.18	72.79

most direct test of the hierarchical features hypothesis stated in §1. Figure 5 presents per-superclass accuracy for all models alongside the depth advantage (M6 – M1) per superclass.

Direct hypothesis test: depth advantage in hierarchical categories. M6 outperforms M1 in 18 of 20 superclasses, with the largest margins in reptiles (+10.8%), people (+9.0%), large herbivores/omnivores (+8.4%), household furniture (+8.2%), and household electrical devices (+7.6%). These are precisely the categories that demand fine-grained intra-class discrimination — distinguishing snake from lizard, or chair from couch — where hierarchical feature composition across multiple abstraction levels provides the greatest benefit over flat, wide representations. This pattern constitutes direct empirical support for the first premise of our hypothesis: *depth enables the compositional hierarchies needed for fine-grained discrimination*, and this advantage is observably concentrated in the semantic categories where such hierarchies matter most.

Direct hypothesis test: width advantage in texture-

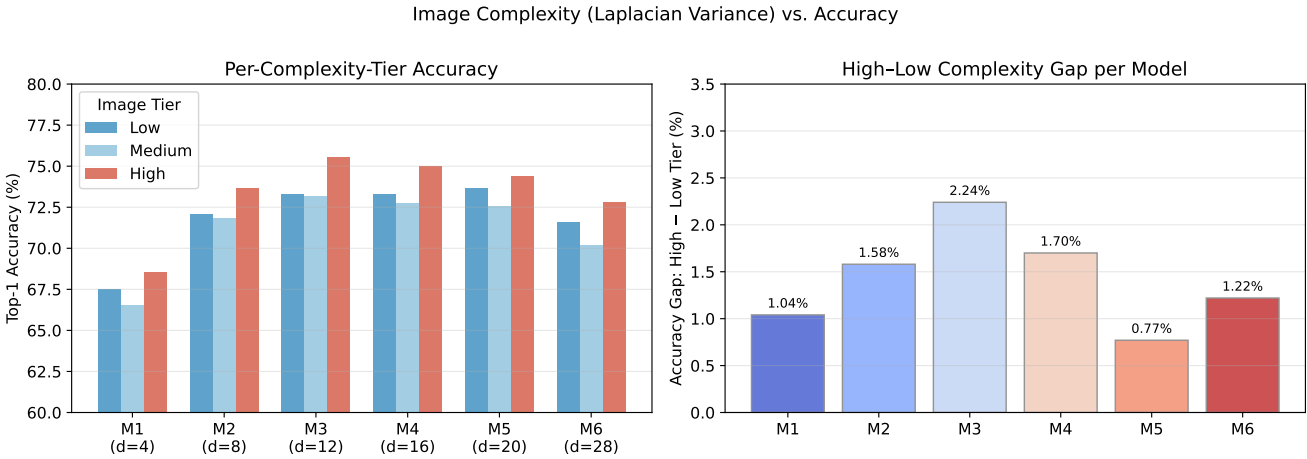


Figure 4. **Left:** Per-complexity-tier accuracy for all models. Tiers are defined by Laplacian variance of the raw test images. **Right:** High-Low accuracy gap per model. The small, non-monotonic gaps indicate that image edge density does not meaningfully modulate the depth-width preference.

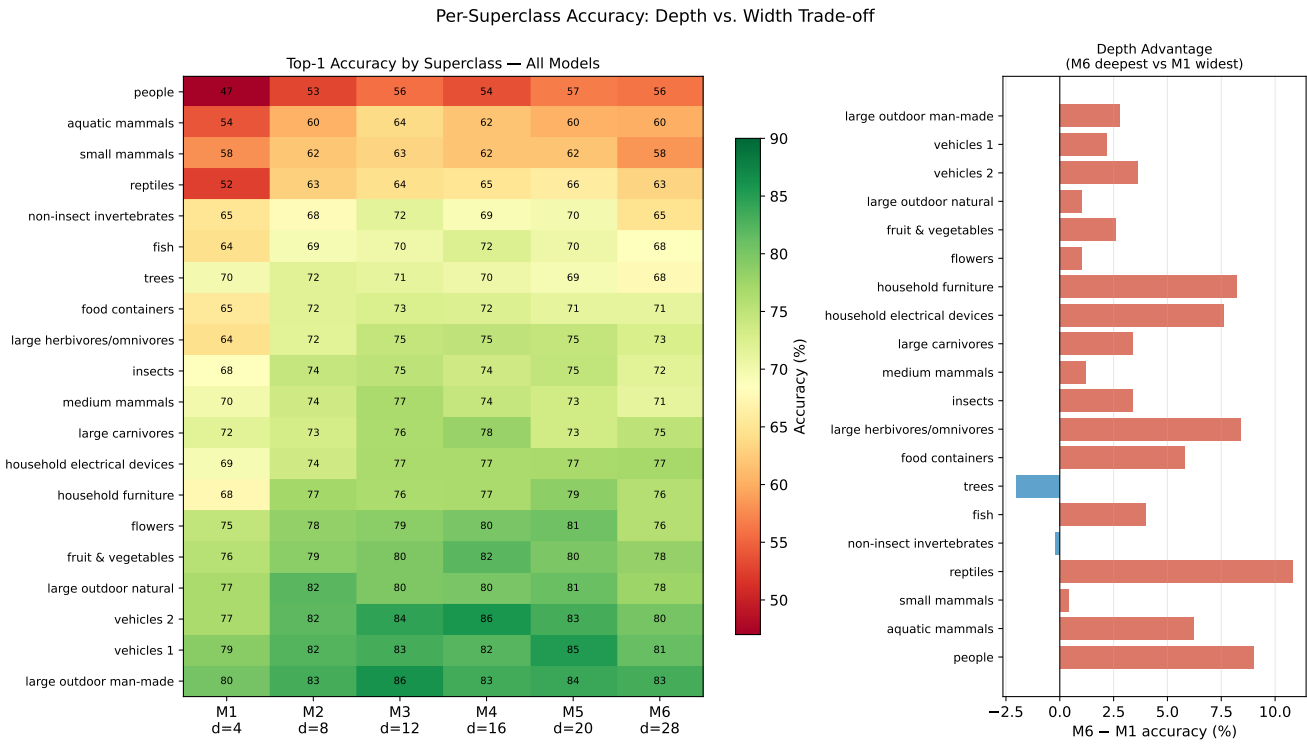


Figure 5. **Left:** Per-superclass top-1 accuracy (%) for all six models, sorted by average difficulty (hardest at top). **Right:** M6 – M1 accuracy difference per superclass. Red bars indicate categories where depth (M6) outperforms width (M1); blue bars indicate the reverse.

dominated categories. The two exceptions where M1 matches or outperforms M6 are trees (–2.0% for M6 relative to M1) and non-insect invertebrates (–0.2%). Trees are characterised by distinctive large-scale colour and texture signatures that can be captured by a single broad convolutional sweep rather than through progressive hierarchical abstraction. This supports the second premise of our

hypothesis: *width encodes richer co-occurrence representations at each level*, providing a relative advantage in categories where large-scale feature statistics dominate.

Why the balanced model M3 remains superior. Despite M6’s broad depth advantage over M1, M3 consistently outperforms M6 across most superclasses, with the largest margins in non-insect invertebrates (+6.8%), medium mam-

mals (+5.4%), and small mammals (+4.8%). For the hardest categories — those requiring the finest intra-class discrimination — M6’s 78-channel width is insufficient even when paired with 28 layers of depth. This demonstrates a *channel capacity floor*: both depth and width are jointly necessary; prioritising one at the extreme expense of the other undermines performance regardless of which hypothesis premise it targets. The optimal configuration, M3, satisfies both premises simultaneously — sufficient depth for hierarchical composition and sufficient width for within-level representational richness — which is precisely what the hypothesis predicts an optimal ratio should achieve.

Hardest categories. Across all models, the lowest accuracy superclasses are people (avg. 53.7%), aquatic mammals (60.1%), small mammals (60.9%), and reptiles (62.3%). These categories involve high intra-class visual similarity and fine-grained inter-instance differences that challenge all six architectures, suggesting that both more depth *and* more width — i.e., a larger parameter budget — would be needed to substantially improve performance on these categories.

6. Discussion and Conclusion

6.1. Discussion

Our results support a unified interpretation: neither extreme depth nor extreme width is individually optimal for CIFAR-100 at $\sim 1.5\text{M}$ parameters. Balanced configurations that allocate sufficient capacity to both the depth dimension (for hierarchical feature building) and the width dimension (for within-level representational richness) achieve the best generalisation.

A practical implication is that for plain CNN design under resource constraints at this scale, a depth-to-width ratio of approximately 12 : 122 (≈ 0.098 layers per channel) represents a well-founded starting point, with the plateau effect between M3 and M5 providing robustness to moderate deviations.

Why does the inverted-U occur? The rising segment (M1 $\rightarrow$ M3) reflects increasing hierarchical capacity: more layers enable more compositional feature representations, directly benefiting fine-grained classification. The falling segment (M3 $\rightarrow$ M6) reflects a capacity-gradient tension: as depth increases beyond the optimum, channel width must decrease to maintain the parameter budget, and the resulting narrow networks suffer both from reduced representational breadth at each level and from attenuated gradient signals in early layers.

Limitations. This study uses a plain (non-residual) VGG-style backbone. Skip connections in ResNets substantially improve gradient flow and may shift the optimal depth-width balance, potentially enabling productive use of greater depth. Our conclusions are specific to this architecture family, $\sim 1.5\text{M}$ parameter scale, and 32×32 resolution;

generalisation to larger budgets, higher-resolution datasets, or modern architectures warrants future investigation. Furthermore, all models in this study are trained from scratch; the depth-width interaction in fine-tuning or transfer learning settings may differ substantially.

6.2. Conclusion

We hypothesised that under a fixed parameter budget, an optimal depth-to-width ratio exists, with depth benefiting hierarchical feature composition and width benefiting within-level representational richness. A systematic study of six plain VGG-style CNNs spanning depths 4–28 on CIFAR-100 confirms this hypothesis on multiple levels. The inverted-U accuracy curve peaking at depth 12 (74.01% top-1) with a stable plateau at depths 12–20 directly confirms that an optimal ratio exists and that both extremes underperform. Gradient flow analysis provides the mechanistic explanation: shallow-wide networks lack the hierarchical composition capacity needed for fine-grained classification, while deep-narrow networks suffer from suppressed gradient magnitudes that impair convergence. Semantic analysis offers the most direct evidence for the hypothesis: depth advantages are largest in fine-grained biological and structural categories — exactly where hierarchical feature composition is theoretically most valuable — while width shows a comparative advantage in texture-dominated categories such as trees, consistent with its role in encoding broader co-occurrence statistics. Image edge complexity does not modulate the trade-off at 32×32 resolution, suggesting the effect is semantic rather than pixel-level. Together, these findings provide both empirical confirmation of our hypothesis and mechanistic insight into why the optimal balance works, offering principled guidance for CNN architecture design under resource constraints and motivating future work extending this analysis to residual architectures and larger parameter scales.

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All experimental results, scientific interpretations, and conclusions presented in this paper are the sole responsibility of the authors. The authors have independently verified the accuracy of all cited references, the correctness of all

reported numerical results, and the validity of all analytical claims. No AI-generated content has been accepted without critical evaluation, and the intellectual contributions of the work — including research design, hypothesis formulation, experimental methodology, and interpretation of findings — were carried out by the authors.

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